

Glass Wool

It is an insulating material, obtained from fibre glass arranged into a texture similar to wood. It is produced or rolled in slabs with different thermal and mechanical properties.

Elements of VA Group

- Elements present in this group are ${}^7\text{N}$, ${}^{15}\text{P}$, ${}^{33}\text{As}$, ${}^{51}\text{Sb}$ and ${}^{83}\text{Bi}$.
- Phosphorus is the most abundant element of this family in the earth's crust.
- Nitrogen is an essential component of explosives. It is generally filled in chips packets or electric bulbs to increase their life because of its inert nature.
- Ammonia NH_3 is an alkaline gas, extremely soluble in water and is dried over quicklime.
- Oxides of nitrogen when dissolve in water give oxyacids, e.g. NO_2 when dissolves in water, gives HNO_3 (nitric acid).
- **Nitric acid** is a strong acid and used in the synthesis of fertilisers and explosives.
- Red phosphorus is largely used in match industry and in the manufacture of phosphor bronze-an alloy of Cu, Sn and P.
- The ignition temperature (543K) is much higher than that of white phosphorus (303K). Hence, it does not catch fire easily.

Elements of VIA Group

(Oxygen Family-Chalcogens)

- Elements in this group are ${}^8\text{O}$, ${}^{16}\text{S}$, ${}^{34}\text{Se}$, ${}^{52}\text{Te}$ and ${}^{84}\text{Po}$.
- Most abundant element of the family is oxygen.
- Common oxidation state of oxygen is (-2) . In gaseous form oxygen is a supporter of combustion.
- Oil of vitriol or king of chemicals is H_2SO_4 .
- Sulphuric acid (H_2SO_4) is a dibasic acid and is a dehydrating agent. It is used in making explosives.
- Ozone (O_3) is an allotrope of oxygen. It is diamagnetic gas.
- SO_2 is a bleaching agent. Its bleaching action is due to reduction and is temporary.
- Oxides of sulphur causes acid rain in the atmosphere.

Elements of VIIth Group (Halogens)

- Elements present in this group are ${}^9\text{F}$, ${}^{17}\text{Cl}$, ${}^{35}\text{Br}$, ${}^{53}\text{I}$ and ${}^{85}\text{At}$. Out of these, bromine (Br_2) exists in liquid state and iodine (I_2) in solid state.
- Fluorine 'F' has maximum electronegativity.
- Chlorine is prepared by oxidation of HCl with MnO_2 , KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ etc.
- Chlorine has maximum electron affinity in the halogens.

- Order of reactivity of halogens, $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$.
- Order of electronegativity, $\text{F} > \text{Cl} > \text{Br} > \text{I}$.
- Order of acidic strength, $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$.
- Iodine is a powerful antiseptic and used as a tincture of iodine (a solution of iodine in alcohol).

Elements of VIIIth Group

- Elements present in this group are ${}^{26}\text{Fe}$, ${}^{27}\text{Co}$, ${}^{28}\text{Ni}$ (Ferrous metals)
 ${}^{44}\text{Ru}$, ${}^{45}\text{Rh}$, ${}^{46}\text{Pd}$ (Light platinum metals)
 ${}^{76}\text{Os}$, ${}^{77}\text{Ir}$, ${}^{78}\text{Pt}$ (High platinum metals)
- All of these are *d*-block elements. Pig iron is obtained from blast furnace. It is most impure form of iron. It is very hard, but brittle. It cannot be welded. It contains maximum amount of carbon (2.5% to 4.3%).
- Wrought iron is the purest form of iron. It contains 0.1 to 0.25% carbon. Wrought iron is soft, ductile and malleable, so it can be welded.
- Steel is the most important form of iron. It contains 0.2 to 1.5% carbon.
- Iron is present in haemoglobin (blood) and its oxidation state in haemoglobin is +2.
- Fe_2O_3 is called jeweller's rouge.
- Vitamin B_{12} is also called cobalamine and contains CO^{3+} ion. Its deficiency in diet causes pernicious anaemia.

Elements of Zero Group

- Elements present in this group are ${}^2\text{He}$, ${}^{10}\text{Ne}$, ${}^{18}\text{Ar}$, ${}^{36}\text{Kr}$, ${}^{54}\text{Xe}$ and ${}^{86}\text{Rn}$ which are called noble or inert gases.
- Argon is the most abundant inert gas in air.
- Helium was discovered by Frankland and Lockyer. He-III is abundant on the lunar surface and holds the potential to put an end to the energy crisis of the earth.
- Argon was discovered by Rayleigh and Ramsay.
- Ne, Kr and Xe were discovered by Ramsay and Traverse.
- Radon was discovered by Dorn and is not present in air.
- All of these elements are colourless monoatomic gases.
- Their electronegativity and electron affinity are zero.

Gas	Uses
Helium	With oxygen for respiration in case of deep sea divers, filling balloons and in low temperature applications.
Neon	In advertising signs.
Argon	Light bulbs
Krypton/Xenon	Photographic flash, TV tubes

Some Important Compounds

Common name	Chemical name	Formula	Uses
<i>Aqua-fortis</i>	Nitric acid	HNO_3	It is used especially in the production of fertilisers and explosives and rocket fuels.
<i>Aqua-regia</i>	Nitric acid + hydrochloric acid	Conc. HNO_3 + conc. HCl (in 1 : 3 ratio)	It is used for testing metals and dissolving platinum and gold.
<i>Baking soda</i>	Sodium bicarbonate	NaHCO_3	It is sodium bicarbonate that is added to baked goods to make them rise.
<i>Brine</i>	Sodium chloride solution	NaCl solution	The electrolysis of brine is a large-scale process used to manufacture chlorine from salt.
<i>Blue vitriol</i>	Copper sulphate	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	It is used as antiseptic agent, dyes, treatment of copper deficiency etc.
<i>Bone ash</i>	Calcium phosphate	$\text{Ca}_3(\text{PO}_4)_2$	Bone ash can be used alone as an organic fertiliser. It can be treated with sulphuric acid to form a "single superphosphetic" fertiliser which is more water soluble.
<i>Bleaching powder</i>	Calcium oxy-chloride	CaOCl_2	It is used as disinfectant and germicide in the sterilization of drinking water.
<i>Caustic soda</i>	Sodium hydroxide	NaOH	It is used as a cleansing agent and in the manufacturing of washing soda.
<i>Chile salt petre</i>	Sodium nitrate	NaNO_3	It is used in solid propellants, explosives, fertilizers etc.
<i>Dry ice</i>	Solid carbon dioxide	CO_2	It is the solid form of CO_2 . It is used primarily as a cooling agent.
<i>Epsom salt</i>	Heptahydrate magnesium sulphate	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	This is used as an antacid or as a mineral supplement to maintain the body's magnesium balance and laxative.
<i>Foul air</i>	Nitrogen	N_2	It is used in food processing, in purging air conditioning and refrigeration.
<i>Grain alcohol</i>	Ethyl alcohol	$\text{C}_2\text{H}_5\text{OH}$	It is used as a solvent, in the synthesis of other organic chemicals. as an additive to automotive gasoline,
<i>Grape sugar</i>	Dextrose	$\text{C}_6\text{H}_{12}\text{O}_6$	Dextrose is a type of sugar better known today as glucose. It is used for treatment of hypoglycemia.
<i>Gypsum</i>	Calcium sulphate dihydrate	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	It is widely used as fertilizer and the main constituent in many forms of plaster, blackboard chalk and wallboard.
<i>Green vitriol</i>	Ferrous sulphate heptahydrate	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	It is also used in medicine to treat iron deficiency.
<i>Gammexane</i>	Benzene hexachloride (BHC)	$\text{C}_6\text{H}_6\text{Cl}_6$	It is also known as lindane. It is used as an agricultural insecticide and as pharmaceutical treatment for lice and scabies.
<i>Heavy water</i>	Deuterium oxide	D_2O	It is used in certain types of nuclear reactors. It acts as a neutron moderator to slow down neutrons.
<i>Halite or table salt</i>	Sodium chloride	NaCl	It is commonly used as a condiment and food preservative.
<i>Lime stone (pearl)</i>	Calcium carbonate	CaCO_3	It is a chemical or biological sedimentary rock that has many uses in agriculture and industry.
<i>Laughing gas</i>	Nitrous oxide	N_2O	This gas used as anaesthetic, as a propellant in whipped cream cans and as an oxidizing agent in racing cars.
<i>Milk of magnesia</i>	Magnesium hydroxide	$\text{Mg}(\text{OH})_2$	It is used as a laxative to relieve occasional constipation and as an antacid to relieve indigestion.
<i>Milk of lime (slaked lime)</i>	Calcium hydroxide	$\text{Ca}(\text{OH})_2$	It is used as lubricant carrier in wire drawing.
<i>Oil of vitriol</i>	Sulphuric acid	H_2SO_4	It uses in the production of fertilizers, manufacture of chemicals in petroleum refining etc.
<i>Plaster of Paris</i>	Calcium sulphate hemihydrate	$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$	It is used for casts to hold broken limbs in place, modeling casts, sculptures and in plasterboard walls and ceilings etc.
<i>Phosgene</i>	Carbonyl chloride	COCl_2	It is a major industrial chemical used to make chemicals and pesticides.
<i>Paris green</i>	Double salt	$\text{Cu}(\text{AsO}_2)_2 \cdot \text{Cu}(\text{C}_2\text{H}_3\text{O}_2)_2$	It is used as a pigment and insecticide. It is used as preservative also.
<i>Pearl white</i>	Bismuthoxychloride	BiOCl	It is used as a yellow pigment for cosmetics and paints.
<i>Quartz or sand</i>	Silicon dioxide	SiO_2	Common uses of silica (quartz) is the manufacture of glass.
<i>Quick silver</i>	Mercury	Hg	It is the chemical element of mercury. It is used in batteries, fluorescent lights, thermometers etc.
<i>Quicklime</i>	Calcium oxide	CaO	It is used in the petroleum industry as an alkali in biodiesel production.
<i>Red lead</i>	Lead tetraoxide	Pb_3O_4	It is used in the manufacture of batteries, lead glass and rust-proof primer paints.
<i>Stranger gas</i>	Xenon	Xe	Xenon is used in photographic flash lamps, stroboscopic lamps, bactericidal lamps, high intensive arc lamps for motion picture protection etc.
<i>Spirit of salt</i>	Hydrochloric acid	HCl	An old fashioned name for hydrochloric acid. It is used in production of batteries. photoflash bulbs and fireworks etc.
<i>Soda ash</i>	Sodium carbonate	Na_2CO_3	It is used in the manufacturing of glass, detergents, chemicals and other industrial products.

Common name	Chemical name	Formula	Uses
Salt cake	Sodium sulphate	Na_2SO_4	It is used in the manufacture of glass, ceramic glazes, soaps and sodium salts.
Tear gas	Chloropicrin	CCl_3NO_2 (or chloro compounds or NH_3)	It is used as a riot control agent.
Vinegar	Acetic acid	CH_3COOH	It is mainly used as cooking ingredient or in pickling.
Washing soda	Sodium carbonate decahydrate	$\text{Na}_2\text{CO}_3 \cdot 10 \text{H}_2\text{O}$	(i) It is used in manufacturing of glass borax, paper etc. (ii) To remove permanent hardness of water.
Water glass	Sodium silicate	Na_2SiO_3	It is used as coagulant/deflocculant agent in wastewater treatment places.
White vitriol	Zinc sulphate	$\text{ZnSO}_4 \cdot 7 \text{H}_2\text{O}$	It is used for preserving skins and wood, in the electro deposition of zinc etc.

ORGANIC CHEMISTRY

ORGANIC COMPOUNDS

- Chemistry of hydrocarbons and their derivatives are called organic compounds which deal with an organic chemistry.
- Urea** (NH_2CONH_2) was the first organic compound prepared in laboratory. It was prepared by **Wohler**.
- Acetic acid** (CH_3COOH) was the first organic compound synthesised from its elements by **Kolbe**.
- Main sources of organic compounds are plants, animals, coal and petroleum.

HYDROCARBONS

Hydrocarbons are the compounds made up of carbon and hydrogen only. These are of two types:

1. Saturated Hydrocarbons or Paraffins or Alkanes (Contain C—C Bond)

- General formula** $\text{C}_n\text{H}_{2n+2}$ where, $n = 1, 2, 3, \dots$
- All the carbon atoms in alkanes are sp^3 -hybridised. They show chain isomerism.
- Methane, CH_4 is the first and least reactive member of the series. Its source is wet and marshy land. It is also present in the air exhaled by animals whose food contains cellulose.
- Cavities in coal contains 90% methane. It is called **fire-damp**. It is responsible for the explosions occurring in coal mines.
- It is used as a gaseous fuel, for preparing carbon black and in the preparation of variety of organic compounds.

2. Unsaturated Hydrocarbons

Contains >C=C< or $\text{—C}\equiv\text{C—}$ Bond

These further are of two types :

(i) Alkenes or Olefins (>C=C<)

- General formula** C_nH_{2n}
- Type formula** R—CH=CH_2
- Common name** Alkylene, **IUPAC name** Alkenes.

- First member is C_2H_4 , ($\text{CH}_2=\text{CH}_2$) known as ethylene or ethene.
- Ethene is used in the manufacture of plastics, for artificial ripening of fruits (banana, apple and orange), as an anaesthesia etc.

(ii) Alkynes or Acetylenes ($\text{—C}\equiv\text{C—}$)

- General formula** $\text{C}_n\text{H}_{2n-2}$
- Type formula** $\text{R—C}\equiv\text{C—H}$
- Common name** Acetylene, **IUPAC name** Alkynes.
- Acetylene (C_2H_2) is the first member of the series.
- It is used in oxyacetylene torch and for welding in acetylene lamps.
- Alkynes can be distinguished from alkenes by the reaction with Tollen's reagent (ammoniacal silver nitrate).

Aromatic Hydrocarbons

- General formula** $\text{C}_n\text{H}_{2n-6}$
- Common name** Arenes
- Most of these compounds contain benzene ring. e.g. benzene, toluene, naphthalene, etc.
- Benzene (C_6H_6) was first synthesised by Berthelot. It is used in the production of various organic compounds, as a solvent and fuel in motor vehicle when mixed with petrol, etc.

Hydrocarbon Derivatives

1. Haloalkanes

- General formula** $\text{C}_n\text{H}_{2n+1}\text{X}$ (where, $\text{X} = \text{Cl, Br, I}$)

Type formula These are of three types:

Primary alkyl halide, $\text{R—CH}_2\text{—X}$

Secondary alkyl halide, $\text{R}_1\text{—}\overset{\text{R}_2}{\underset{|}{\text{CH}}}\text{—X}$



- **Common name** Alkyl halide;
IUPAC name Haloalkane.
- Trihalo methane are called **haloforms**, e.g. chloroform ($CHCl_3$), iodoform (CHI_3), etc.
- Chloroform oxidises to $COCl_2$ (carbonyl chloride or phosgene gas) in the presence of air and light. So, it is kept in closed dark coloured bottles and mixed with ethyl alcohol which acts as negative catalyst and form non-poisonous, ethyl carbonate with $COCl_2$.
- Chloroform is used as a general anaesthetic.
- Chloropicrin ($CCl_3 \cdot NO_2$) is used as an insecticide and also called **tear gas**. Carbon tetrachloride (CCl_4) is used for extinguishing fire under the name **pyrene**.
- Ethylene dibromide is added to petrol in place of tetraethyl lead to save the environment from lead pollution.
- Chlorofluorocarbons (freons or CFC) are used as refrigerants. They contain carbon, chlorine, hydrogen and fluorine and destroy the ozone layer.

2. Alcohols

- **General formula** $C_nH_{2n+1} \cdot OH$
- **Common name** Alkyl alcohol;
IUPAC name Alkanol.
- Primary alcohol is $R-CH_2OH$, secondary alcohol is R_2CH-OH and tertiary alcohol is R_3C-OH .
- Lucas reagent (conc. HCl + anhydrous $ZnCl_2$) is used to distinguish between primary, *sec* and *tert*-alcohols.
- CH_3OH is produced by the destructive distillation of wood, hence also known as, wood spirit, methyl alcohol, methanol or carbinol. It is used as a solvent for paints and varnishes, in making dyes, drugs, perfumes and also used as motor fuel with petrol.
- The destructive distillation of wood yields wood gas (gaseous), tar (liquid), charcoal (solid residue), methyl alcohol (liquid) and acetic acid (vinegar).
- Various products obtained by the destructive distillation of coal are coal gas ($H_2 + CH_4 + CO$), ammoniacal liquor, coal tar and coke.
- CH_3CH_2OH is ethyl alcohol or ethanol or grain alcohol. It is made unfit for drinking purpose by adding methanol or pyridine. Such alcohol is called **methylated spirit** or **denatured alcohol**.

- Ethanol changes the colour of sodium or potassium dichromate from orange to green. It is also used in the manufacture of paints, varnishes, in making transparent soap in the preparation of chloroform, as a fuel in internal combustion engines. It is used in alcoholic drinks, whisky, wine, beer, etc.

3. Ethers

- **General formula** $C_nH_{2n+2} \cdot O$
- **Common name** Dialkyl ether
- **IUPAC name** Alkoxy alkane
- Ethers behave like Lewis bases.
- Diethyl ether ($C_2H_5OC_2H_5$) is used as an industrial solvent and also as an anaesthesia.

4. Aldehydes

- **General formula** $C_nH_{2n+1}CHO$ or $C_nH_{2n}O$;
- **IUPAC name** Alkanal
- 40% solution of formaldehyde is called **formalin** and used in the preservation of dead animals.
- Benzaldehyde is almond extract.
- Paraldehyde (a carbonyl compound) is used as a hypnotic.
- Eugenol is an active component of clove oil.

5. Ketones

- **General formula** $C_nH_{2n}O$ or $(C_nH_{2n+1})_2O$
- **IUPAC name** Alkanone.
- Ketones containing $-COCH_3$ (methyl ketonic group) gives yellow colour (iodoform) in iodoform test.
- Acetone (CH_3COCH_3) is the first member of this class.
- Acetone is used as nail polish remover. However, now-a-days, acetone free removers are in common use. Its main ingredient is ethyl acetate or butyl acetate.
- Methyl Ethyl Ketone (MEK) is used as paint stripper.
- Aldehydes and ketones having α -hydrogen atom give aldol condensation. Such carbonyl compounds which have α -hydrogen atom undergo condensation in the presence of dilute base to form aldol.

6. Carboxylic Acids

- **General formula** $C_nH_{2n+1} \cdot COOH$ or $C_nH_{2n}O_2$;
IUPAC name Alkanoic acid.
- First member of the carboxylic acid series is formic acid, $HCOOH$, which is present in ant's or bee's sting.
- $HCOOH$ is the strongest acid among the carboxylic acids.
- CH_3COOH is the ethanoic or acetic acid (vinegar).
- Vinegar contains 6-10 % acetic acid. Acetic acid is used in the manufacture of rubber from latex and casein from milk. It is used for coagulation.
- Lactic acid is α -hydroxy propanoic acid. It is present in milk and provides it sour taste.

- Citric acid is a hydroxy tricarboxylic acid. It is present in citrus fruits.
- Oxalic acid removes rust stains.
- Salicylic acid is used to prepare aspirin (*o*-acetyl salicylic acid).

7. Esters

- **General formula** $C_nH_{2n+1}COOR$ or $C_nH_{2n}O_2$
- **IUPAC name** Alkyl alkanoate.

- These are used in making artificial perfumes, flavours and essence used in cold drinks, ice-creams, etc.
- Sodium benzoate is used as a food preservative.
- These have characteristic sweet fruity smell.

Ester	Flavour	Ester	Flavour
Amyl acetate	Banana	Isoamyl valerate	Apple
Octyl acetate	Orange	Methyl butyrate	Pineapple

MAN-MADE MATERIALS

SOAPS

Soaps are the sodium or potassium salts of long chain carboxylic (fatty) acids ($RCOONa$) e.g. sodium palmitate ($C_{15}H_{31}COONa$), sodium stearate ($C_{17}H_{35}COONa$) and sodium oleate ($C_{17}H_{33}COONa$) etc. Animal fat or vegetable oil, sodium hydroxide, sodium chloride act as the raw materials for the manufacture of ordinary soap. The process of making soap is called **saponification**.

- Soaps do not form lather with hard water due to the formation of scum.
- When the medium is acidic, soaps form insoluble long chain fatty acids and hence, lost their cleansing action.

DETERGENTS

- These are also called “soap-less” soaps as they do not contain any soap.
- These are the long chain benzene sulphonic acid or the sodium salt of the long chain alkyl hydrogen sulphate.
- Long chain hydrocarbons, sulphuric acid and sodium hydroxide are the raw materials for the manufacturing of detergents.
- These are non-biodegradable and cause water pollution. However, straight chain detergents are biodegradable.
- They form lather with soft as well as hard water.

Polymers

A polymer is a compound of high molecular weight formed by the union of a large number of molecules of one or two types of low molecular weight (known as monomers) and the process involving the formation of a polymer is called **polymerisation**.

Types of Polymers

On the basis of their origin, the polymers are classified in the following manner:

1. **Natural Polymers** These are found in nature, e.g. cellulose, starch, rubber, wool, silk etc.
2. **Synthetic Polymers** These are prepared in the laboratory by synthetic means, e.g. polythene, nylon, orlon, dacron, etc.

3. **Semi-synthetic Polymers** These are synthesised by man from natural substances, e.g. rayon (a polymer of cellulose nitrate) is also called artificial silk due to its silk like appearance.

On the basis of intermolecular forces, these are classified as:

1. **Elastomers** In these polymers, the polymer chains are held together by weak van der Waals' forces, e.g. vulcanised rubber.
2. **Fibres** In this type, polymer chains are attached with one another through H-bonds, e.g. nylon-6,6.
3. **Thermoplastics** In these, the intermolecular forces are intermediate of elastomers and fibres. These are soften on heating and become hard on cooling, e.g. polystyrene, polythene, PVC, etc.
4. **Thermosetting** These are highly cross-linked, hard, non-fusible and insoluble polymers. These can only be moulded into desired shape on heating once. e.g. bakelite (phenol-formaldehyde resin), melamine, etc.

Some Important Synthetic Polymers

- **Polyethylene** (polythene) The monomer units are ethylene molecules. It is frequently used in making coats, milk cartons and electrical insulation.
- **Polystyrene** The monomer units are styrene molecules. It is a white thermoplastic material and is used for making toys, combs, lining material for refrigerators and TV cabinets.
- **Teflon** (Polytetrafluoroethylene) The monomer unit is tetrafluoroethylene molecule. It is very tough material. It is a bad conductor of electricity and is used in coating utensils, making seals, baskets, pipes, flooring, etc.
- **Polyvinyl Chloride (PVC)** The monomer units are vinyl chloride molecules. PVC is a hard, horny material. It is resistant to chemicals as well as heat. It is used for making raincoats, hand bags, electrical insulators and floor covering.
- **Nylon or Nylon-6, 6** It was first fully synthetic fibre and has good elasticity, low water absorption and wrinkle resistant. It has also high tensile strength. It is used in making fishing nets, tyre cord, parachute fabrics, ropes, etc.